

# Yatin Dandi

PH.D STUDENT · SCHOOL OF COMPUTER AND COMMUNICATION SCIENCES

EPFL, Lausanne

☎ (+41)782255109 | ✉ yatin.dandi@epfl.ch | 🌐 yatindandi.github.io | 📷 yatindandi

## Education

### École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

PH.D, COMPUTER SCIENCE

September, 2022 - Present

- **Advisors:** Prof. Lenka Zdeborová, Statistical Physics of Computation Laboratory (SPOC), and Prof. Florent Krzakala, Information, Learning and Physics Laboratory (IdePHICS).
- Program associate at the Simons Laufer Mathematical Sciences Institute (SLMath), Berkeley, Probability and Statistics of Discrete Structures, February-May 2025.

### Indian Institute of Technology Kanpur

Kanpur, India

DUAL DEGREE (BTECH-MTECH), COMPUTER SCIENCE AND ENGINEERING

July, 2017 - August 2022

- **Thesis title:** Tackling Heterogeneity in Collaborative Machine Learning, advisors: Prof. Martin Jaggi (EPFL) and Prof. Piyush Rai (IIT Kanpur).
- **Cumulative Performance Index (CPI) :** 9.6/10.0, Master's CPI: 10.0/10.0
- Exchange and visiting student at Ecole polytechnique fédérale de Lausanne (EPFL)

## Honors & Awards

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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 2026      | <b>EPFL AI Center PhD Fellowship</b> , Fellowship supporting the salary of outstanding PhD candidates working on “high-impact AI research”.       | Switzerland   |
| 2026      | <b>Honorable Mention, Jane Street Graduate Research Fellowship</b> ,                                                                              |               |
| 2024      | <b>EPFL Doc.Mobility grant</b> , Grant awarded for conducting academic research at the Massachusetts Institute of Technology, USA.                | Switzerland   |
| 2022      | <b>EDIC Fellowship, EPFL</b> , Fellowship for the first year of doctoral studies.                                                                 | Switzerland   |
| 2017      | <b>Aditya Birla Scholarship</b> , Awarded to 15 students across all IITs                                                                          | India         |
| 2018,2019 | <b>Academic Excellence Award, IIT Kanpur</b> , Awarded (twice) for exceptional performance                                                        | India         |
| 2016      | <b>KVPY Scholarship, 2016</b> , Indian Institute of Science                                                                                       | India         |
| 2015      | <b>NTSE Scholarship</b> , Government of India                                                                                                     | India         |
| 2017      | <b>All India Rank 135 (out of 172024 students)</b> , JEE Advanced 2017, second stage of admission process to IITs having 1.186 million applicants | India         |
| 2016      | <b>Selected for Indian National Physics and Chemistry Olympiads</b> , HBCSE                                                                       | Mumbai, India |

## Publications and Preprints

### Deep Learning as Neural Low-Degree Filtering: A Spectral Theory of Hierarchical Feature Learning

YATIN DANDI, MATTEO VILUCCHIO, LUCA ARNABOLDI, HUGO TABANELLI, FLORENT KRZAKALA

HiLD workshop, ICML 2026, Arxiv: <https://arxiv.org/abs/2605.13612>

### Scaling Laws from Sequential Feature Recovery: A Solvable Hierarchical Model

ARIE WORTSMAN-ZURICH, HUGO TABANELLI, YATIN DANDI, FLORENT KRZAKALA, BRUNO LOUREIRO

HiLD workshop, ICML 2026, Arxiv: <https://arxiv.org/abs/2605.14567>

### Rigorous Asymptotics for First-Order Algorithms Through the Dynamical Cavity Method

YATIN DANDI, DAVID GAMARNIK, FRANCISCO PERNICE, LENKA ZDEBOROVÁ

Conference on Learning Theory (COLT), 2026, Arxiv: <https://arxiv.org/abs/2603.14573>

### Deep Learning of Compositional Targets with Hierarchical Spectral Methods

HUGO TABANELLI, YATIN DANDI, LUCA PESCE, FLORENT KRZAKALA

International Conference on Machine Learning (ICML), 2026, Arxiv: <https://arxiv.org/abs/2602.10867>

### Minimum Number of Monochromatic Subgraphs of a Random Graph

YATIN DANDI, DAVID GAMARNIK, HAODONG ZHU

Arxiv: <https://arxiv.org/abs/2602.03774>

### Provable Learning of Random Hierarchy Models and Hierarchical Shallow-to-Deep Chaining

YUNWEI REN, YATIN DANDI, FLORENT KRZAKALA, JASON D. LEE

Conference on Learning Theory (COLT), 2026, Arxiv: <https://arxiv.org/abs/2601.19756>

JULY 28, 2026

## **The Computational Advantage of Depth: Learning High-Dimensional Hierarchical Functions with Gradient Descent**

YATIN DANDI, LUCA PESCE, LENKA ZDEBOROVÁ, FLORENT KRZAKALA

**Spotlight**, 39th Conference on Neural Information Processing Systems (NeurIPS) 2025, Arxiv: <https://arxiv.org/abs/2502.13961>

## **Fundamental limits of learning in sequence multi-index models and deep attention networks: High-dimensional asymptotics and sharp thresholds**

EMANUELE TROIANI, HUGO CUI, YATIN DANDI, FLORENT KRZAKALA, LENKA ZDEBOROVÁ

International Conference on Machine Learning (2025), Arxiv: <https://arxiv.org/abs/2502.00901>

## **A random matrix theory perspective on the spectrum of learned features and asymptotic generalization capabilities.**

YATIN DANDI, LUCA PESCE, HUGO CUI, FLORENT KRZAKALA, YUE M. LU, BRUNO LOUREIRO

**Oral**, Artificial Intelligence and Statistics (AISTATS) 2025, Arxiv: <https://arxiv.org/abs/2410.18938>

## **Fundamental limits of weak learnability in high-dimensional multi-index models**

EMANUELE TROIANI, YATIN DANDI, LEONARDO DEFILIPPIS, LENKA ZDEBOROVÁ, BRUNO LOUREIRO, FLORENT KRZAKALA

Artificial Intelligence and Statistics (AISTATS) 2025, Arxiv: <https://arxiv.org/abs/2405.15480>

## **Asymptotics of feature learning in two-layer networks after one gradient-step**

HUGO CUI, LUCA PESCE, YATIN DANDI, FLORENT KRZAKALA, YUE M. LU, LENKA ZDEBOROVÁ, BRUNO LOUREIRO

**Spotlight**, International Conference on Machine Learning, 2024, Arxiv: <https://arxiv.org/abs/2402.04980>

## **The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks: Breaking the Curse of Information and Leap Exponents**

YATIN DANDI, EMANUELE TROIANI, LUCA ARNABOLDI, LUCA PESCE, LENKA ZDEBOROVÁ, FLORENT KRZAKALA

International Conference on Machine Learning, 2024, Arxiv: <https://arxiv.org/abs/2402.03220>

## **Online Learning and Information Exponents: The Importance of Batch size & Time/Complexity Tradeoffs**

LUCA ARNABOLDI, YATIN DANDI, FLORENT KRZAKALA, BRUNO LOUREIRO, LUCA PESCE, LUDOVIC STEPHAN

International Conference on Machine Learning, 2024, Arxiv: <https://arxiv.org/abs/2406.02157>

## **Asymptotics of Non-Convex Generalized Linear Models in High-Dimensions: A proof of the replica formula**

MATTEO VILUCCHIO, YATIN DANDI, CEDRIC GERBELOT, FLORENT KRZAKALA

Preprint, Arxiv: <https://arxiv.org/abs/2502.20003>

## **Repetita iuvant: Data repetition allows sgd to learn high-dimensional multi-index functions**

LUCA ARNABOLDI, YATIN DANDI, FLORENT KRZAKALA, LUCA PESCE, LUDOVIC STEPHAN

Preprint, Arxiv: <https://arxiv.org/abs/2405.15459>

## **Sampling with flows, diffusion and autoregressive neural networks: A spin-glass perspective**

DAVIDE GHIO, YATIN DANDI, FLORENT KRZAKALA, AND LENKA ZDEBOROVÁ

Proceedings of the National Academy of Sciences, Vol. 121 | No. 27 July 2, 2024, NeurIPS 2023 Workshop on Diffusion Models, Arxiv: <https://arxiv.org/abs/2308.14085>

## **How Two-Layer Neural Networks Learn, One (Giant) Step at a Time**

YATIN DANDI, FLORENT KRZAKALA, BRUNO LOUREIRO, LUCA PESCE, LUDOVIC STEPHAN

Journal of Machine Learning Research (JMLR) **Invited talk**, NeurIPS 2023 Workshop: Mathematics of Modern Machine Learning (M3L), Arxiv: <https://arxiv.org/abs/2305.18270>

## **Universality laws for Gaussian mixtures in generalized linear models**

YATIN DANDI\*, LUDOVIC STEPHAN\*, FLORENT KRZAKALA, BRUNO LOUREIRO, LENKA ZDEBOROVÁ (\* DENOTES EQUAL CONTRIBUTION)

Conference on Neural Information Processing Systems (NeurIPS), 2023, Arxiv: <https://arxiv.org/abs/2302.08933>

## **Maximally-stable Local Optima in Random Graphs and Spin Glasses: Phase Transitions and Universality**

YATIN DANDI, DAVID GAMARNIK, LENKA ZDEBOROVÁ

Presented at the meeting on “Random Graphs: Combinatorics, Complex Networks and Disordered Systems” at the Oberwolfach Research Institute for Mathematics in March. Arxiv: <https://arxiv.org/abs/2305.03591>

## **Data-heterogeneity-aware Mixing for Decentralized Learning**

YATIN DANDI, ANASTASIA KOLOSKOVA, MARTIN JAGGI, SEBASTIAN STICH

NeurIPS 2022 Workshop: Optimization for Machine Learning, Arxiv: <https://arxiv.org/abs/2204.06477>

## Implicit Gradient Alignment in Distributed and Federated Learning

YATIN DANDI\*, LUIS BARBA\*, MARTIN JAGGI (\* DENOTES EQUAL CONTRIBUTION)

Proceedings of the AAAI Conference on Artificial Intelligence (Preliminary version also accepted at the FL-ICML Workshop, 2021), Arxiv:

<https://arxiv.org/abs/2106.13897>

## Generalized Adversarially Learned Inference

YATIN DANDI, HOMANGA BHARADHWAJ, ABHISHEK KUMAR, PIYUSH RAI

Proceedings of the AAAI Conference on Artificial Intelligence (Preliminary version also accepted at the NeurIPS 2020 Workshop: Self-Supervised Learning – Theory and Practice), Arxiv: <https://arxiv.org/abs/2006.08089>

## Jointly Trained Image and Video Generation using Residual Vectors

YATIN DANDI, ANIKET DAS, SOUMYE SINGHAL, VINAY P. NAMBOODIRI, PIYUSH RAI

2020 Winter Conference on Applications of Computer Vision (WACV '20), Arxiv: <https://arxiv.org/abs/1912.07991>

## Model-Agnostic Learning to Meta-Learn

ARNOUT DEVOS\*, YATIN DANDI\* (\* DENOTES EQUAL CONTRIBUTION)

Proceedings of Machine Learning Research (PMLR) (Preliminary version accepted at the Pre-registration workshop, NeurIPS (2020)), Arxiv:

<https://arxiv.org/abs/2012.02684>

## NeurInt-Learning Interpolation by Neural ODEs

AVINANDAN BOSE\*, ANIKET DAS\*, YATIN DANDI, PIYUSH RAI

**Spotlight**, The Symbiosis of Deep Learning and Differential Equations: DLDE Workshop, NeurIPS 2021. Arxiv:

<https://arxiv.org/abs/2111.04123>

## Understanding Layer-wise Contributions in Deep Neural Networks through Spectral Analysis

YATIN DANDI, ARTHUR JACOT

Preprint, Arxiv: <https://arxiv.org/abs/2111.03972>

## Internships

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### Optimization, Sampling, and Trajectory Inference algorithms

MILA

INTERNSHIP UNDER PROF. SIMON LACOSTE-JULIEN AND PROF. YOSHUA BENGIO

June 2022 - August 2022

- Proposed and analyzed a general theoretical framework for proving convergence rates of min-max optimization algorithms.
- Proved results related to the saddle-point Frank Wolfe algorithm.
- Worked on the analysis of sampling and trajectory Inference Algorithms motivated by applications in single-cell Biology.

### New York Office, IIT Kanpur

IIT Kanpur

MACHINE LEARNING FOR LARGE SCALE LOGISTICS PLATFORM, UNDER PROF. MANINDRA AGARWAL

May 2018 - July 2018

- Implemented a state of the art algorithm for online collaborative filtering based on Fast Matrix Factorization for Online Recommendation with Implicit Feedback (He et al.) using Numpy and improved the model with sentiment and frequency dependent weighting schemes.
- Used Kafka for real-time data processing and simulated interactions using locust.
- Implemented a recommender system based on deep autoencoders and compared the results with other models using metrics such as hit ratio.
- Implemented a Bidirectional LSTM model using Keras for sentiment analysis of user comments.
- Trained the Latent Dirichlet allocation model on Wikipedia articles for automatic extraction of topics.

## Talks, Seminars, and Presentations

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- Delivered a talk at the HILD workshop, Banff, Canada, 2026.
- Delivered a talk at Youth in High-Dimensions, Trieste, 2026.
- Delivered a talk at the ProbAI Scaling Laws Workshop, Warwick, 2026.
- Delivered a talk at the Physics of ML kickoff workshop, Stanford University, California, USA, 2025.
- Delivered a talk at “Statistical Physics and Machine Learning: Moving Forward”, Cargese, Corsica, France, 2025.
- Delivered a talk at The Third Joint SIAM/CAIMS Annual Meetings, Montreal, Canada, 2025.
- Delivered a talk at the YEP workshop, Eindhoven, Netherlands, 2025
- Delivered a talk at the ML Theory reading group, UC Berkeley, California, USA, 2025.
- Delivered a talk at the Graduate Student Seminar at the Simons Laufer Mathematical Sciences Institute (SLMath), Berkeley, California, USA, 2025
- Delivered a talk at the 2025 Joint Mathematics Meetings (JMM 2025), Seattle, USA, 2025
- Delivered a talk at the Leman-Th 2024 Workshop on Machine Learning & Neural Network Theory at EPFL, Switzerland, 2024
- Delivered a talk at the “Probability Theory Seminar” at EPFL, Switzerland, 2023.
- Delivered a talk at “Statistical Physics and Machine Learning Back Together Again”, Cargese, Corsica, France, 2023.

- Delivered a talk during the meeting “Random Graphs: Combinatorics, Complex Networks and Disordered Systems” at the Oberwolfach Research Institute for Mathematics in March, 2023.
- Presented a poster at “High Dimensional Statistics and Random Matrices” in Porquerolles, France, 2023.
- Presented a poster and participated in the Princeton Machine Learning Theory Summer School, Princeton University, USA, 2023.
- Presented a poster and participated in the Summer school on Statistical Physics and Machine learning at Ecole de Physique des Houches, France, 2022.
- Presented a poster at the Workshop on Spin Glasses, Les Diablerets, Switzerland, 2022.

## Reviewing

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Neurips(2025), ICML(2025), Neurips(2024), ICML(2024), SIAM Journal on Mathematics of Data Science (2023), TMLR(2023), Neurips(2023), ICML(2023), ICLR(2023), AISTATS (2022)

## Teaching and Supervision

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- **Teaching Assistant:** Statistical physics III (EPFL), Statistical physics for Optimization and Learning (EPFL), Statistical physics of computation (EPFL), Elements of statistics for data science (EPFL), Mathematics for Computer Science - 2, 3 (IITK), Introduction to Machine Learning (IITK)
- **Supervisor - Master’s thesis project:** Hardness of Learning through Gradient Descent in Convolutional Neural Networks, Mia Zosso, 2024

## Programming Skills

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### Programming Languages

C, C++, Python, Javascript, MATLAB, Julia, Haskell

### Libraries and frameworks

Pytorch, Tensorflow, Jax, Scikit-Learn, Numpy, Pandas, Pillow, Keras

### Web

Flask, HTML, CSS, jQuery, MySQL

### Utilities

Linux shell utilities, LaTeX, Git, Docker